GUHNOO YUN

Ph.D. Candidate
Computer Science and Engineering, Korea University
Student Researcher
Intelligence and Interaction Research Center, KIST
Seoul, Republic of Korea

cheeryun@gmail.com
doranlyong.github.io
Google Scholar
GitHub
LinkedIn

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RESEARCH STATEMENT

Analyzing and developing the inner workings of AI perception models to improve how machines see and understand the real world. AI perception models often work well before anyone knows why. My research explores the underlying mechanisms, often by treating each core component inside a model as a filter and analyzing what it keeps or removes as information passes through it, from broad patterns to fine details. I then turn that understanding into better designs.

- Analyzed convolution and attention as spectral filters, and designed new components that balance fine details and broad patterns. Then grounded them in theory and outperformed leading vision architectures. [1, 4]
- Viewed linear attention, whose cost grows only in proportion to input size, as a fixed-size memory, and used a filter to decide what that memory keeps. The result is a general-purpose operation that works across image classification, object detection, segmentation, and image generation. [11]
- Built perception systems for healthcare robots and daily life, some combining cameras with brain waves and body motion. They detect falls of older adults and support robot services in hospital isolation wards. [2, 3, 5, 6, 8, 9]

Looking ahead, I aim to unite the different mathematical views of AI perception models, such as filters and memories, into one understanding that brings machines closer to human intelligence. My work already builds on both of these views. With this understanding, I will design models that combine video, sound, and body signals, and that run well on the chips inside real machines such as humanoid robots. The goal is machines that understand the real world, built on designs we understand.

EDUCATION

Feb 2027 (expected) **Ph.D. in Computer Science and Engineering**
Korea University Seoul, Korea
Dissertation: *Spectral Analysis and Adaptive Filtering for Visual Perception*

2018 **M.S. in Mechatronics**
Gwangju Institute of Science and Technology (GIST) Gwangju, Korea
Thesis: *Removal of Jitter Noise using a Modified Kalman Filter in 3D Shape Recovery from Focus*

2016 **B.S. in Control and Measurement Engineering**
Gyeongsang National University (GNU) Jinju, Korea

RESEARCH EXPERIENCE

Since 2019 **Student Researcher**
Intelligence and Interaction Research Center, **Korea Institute of Science and Technology (KIST)**
Supervisor: Dr. Dong Hwan Kim

Linear Attention for Generative and Vision Transformers (SALAFormer; in

preparation [11])

- Designed **SALA** (Spectral-Adaptive Linear Attention) as closed-form associative memory: each head solves a small ridge regression for what to store in its fixed-size memory, instead of summing raw key–value products, and a learned spectral filter on the key Gram matrix allocates that memory, at linear $O(N)$ inference cost.
- **Image generation:** lower FID than MHLA (ICLR 2026), a recent linear-attention design for diffusion transformers, at both model sizes: 59.80 → **57.65** (DiT-S/2) and 37.47 → **36.54** (DiT-B/2) vs. its published results (class-conditional ImageNet 256×256, no guidance).
- **Image classification:** SALA in a vision backbone (SALAFormer) reaches **77.2%** / **81.8%** ImageNet-1K top-1 at DeiT-Tiny/Small size, above the 76.5% / 81.6% reported for ViT^3 (CVPR 2026) with the same recipe.

Spectral Analysis of Convolution and Attention for Vision Backbones

- Developed **SPANetV2**: compared what convolution and self-attention pass or suppress in one graph-spectral framework, showing that window size shapes their frequency responses, and turned this into a mixer with multi-scale kernels and spectral re-scaling; the backbone outperforms state-of-the-art models on ImageNet-1K, COCO, and ADE20K. [1]
- Proposed **SPANet**: prior work found that self-attention acts as a low-pass filter, while convolution retains high frequencies, and that stronger high-pass filtering helps self-attention. SPANet showed that stronger low-pass filtering also helps convolution, and that a token mixer *balancing* high and low frequencies improves accuracy. [4]

Applied Vision for Robotics, Healthcare & 3D Shape Recovery

- Fall detection of older adults by a healthcare robot [6]; robot-assisted services in isolation wards [5]; daily-life arousal monitoring and brain–computer interface decoding with egocentric vision and EEG [2, 3]; a view-consistent evaluation metric for multi-view panoptic segmentation [10].
- Shape from focus, a follow-up to the M.S. thesis: 3D shape from few images with Gaussian-process regression [7].

2018 – 2019 **Intern Researcher**
Center for Intelligent Robotics, **KIST**

- Perception for healthcare robots: detecting when people with dementia leave home, to keep them from going missing [9]; accelerometer–depth fall detection for older adults [8], later extended with LSTMs [6].

PUBLICATIONS

Published / Accepted

[1] Guhnoo Yun, J. Yoo, K. Kim, J. Lee, P.H. Seo, D.H. Kim, **Spectral-Adaptive Modulation Networks for Visual Perception.** *IEEE Trans. Pattern Anal. Mach. Intell. (TPAMI)* 48(10), 11511–11525 (2026). [doi: 10.1109/TPAMI.2026.3690455] [arXiv] [code]

[2] W. Hyung, M. Kim, Y.-S. Kim, J. Lee, B. Ko, Guhnoo Yun, D.H. Kim, C.-H. Im, **Context-Aware Daily-Life Arousal Monitoring Using Egocentric Vision and EEG.** *Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)* (2026). [program]

[3] Y.-S. Kim, M. Kim, W. Hyung, B. Ko, Guhnoo Yun, D.H. Kim, C.-H. Im, **PeBCI: Vision-Guided Brain-Computer Interface for Context-Aware Neural Decoding in Daily Life.** *Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)* (2026). [program]

[4] Guhnoo Yun, J. Yoo, K. Kim, J. Lee, D.H. Kim, **SPANet: Frequency-balancing Token Mixer using Spectral Pooling Aggregation Modulation.** *IEEE/CVF Int. Conf. on Computer Vision (ICCV)*, 6090–6101 (2023). [doi: 10.1109/ICCV51070.2023.00562] [CVF] [arXiv] [project] [code]

[5] Y. Kwon, S. Shin, K. Yang, S. Park, S. Shin, H. Jeon, K. Kim, Guhnoo Yun, …, Y. Lim, **Heterogeneous Robot-Assisted Services in Isolation Wards: A System Development and Usability Study.** *IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 8069–8076 (2023). [doi: 10.1109/IROS55552.2023.10341857]

[6] K. Kim, Guhnoo Yun, S.-K. Park, D.H. Kim, **Efficient Fall Detection for a Healthcare Robot System Based on 3-Axis Accelerometer and Depth Sensor Fusion with LSTM Networks.** *Int. Conf. on Pattern Recognition (ICPR)*, 2207–2212 (2022). [doi: 10.1109/ICPR56361.2022.9956418]

[7] H.-S. Jang, Guhnoo Yun, M.T. Mahmood, M.-K. Kang, **Optimal Sampling for Shape from Focus by Using Gaussian Process Regression.** *IEEE Int. Conf. on Consumer Electronics (ICCE)*, 1–4 (2020). [doi: 10.1109/ICCE46568.2020.9043150]

[8] K. Kim, Guhnoo Yun, S.-K. Park, D.H. Kim, **Fall Detection for the Elderly Based on 3-Axis Accelerometer and Depth Sensor Fusion with Random Forest Classifier.** *Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 4611–4614 (2019). [doi: 10.1109/EMBC.2019.8856698]

[9] Guhnoo Yun, K. Kim, S.-K. Park, D.H. Kim, **A Monitoring System to Support Home Health Care for the Elderly with Dementia by Detecting Going Out Activities Based on RGB-D Sensors.** *Int. Conf. on Ubiquitous Robots (UR)*, 71–76 (2019). [doi: 10.1109/URAI.2019.8768598]

Under Review

[10] Y. Lee, B. Ko, Guhnoo Yun, D.H. Kim, **Every View Counts: View-Consistent Panoptic Quality for Multi-view Panoptic Segmentation.** *arXiv preprint* (2026). [arXiv: 2610.05911]

In Preparation

[11] Guhnoo Yun et al., **SALAFormer: Spectral-Adaptive Linear Attention as Closed-Form Associative Memory.**

PATENTS

2025 **Frequency-Balancing Token Mixer and Its Backbone Model**
Korean Patent No. 10-2815184 (registered May 27, 2025)

2022 **System and Method for Determining Whether a Resident Has Left Home**
Korean Patent No. 10-2374057 (registered Mar 8, 2022)

AWARDS & SCHOLARSHIPS

2026 **Outstanding Student Researcher Scholarship**
Korea Institute of Science and Technology (KIST)

2023 **Outstanding Technology Award**
Korea Institute of Science and Technology (KIST)

2020 **Presidential Award** 2020 International Robot Contest—Humanoid Robot Sports (Deep Learning Algorithm)
Ministry of the Interior and Safety

2016 – 2017 **Graduate Scholarship**
Gwangju Institute of Science and Technology (GIST)

2013 – 2015 **Academic Excellence Scholarship**
Gyeongsang National University (GNU)

INVITED TALKS

2026 **2026 Inter-Ministerial Public Technology Transfer and Commercialization Roadshow**
Presentation at the Science and Technology Convention Center, Seoul

2021 **"Recent Research Trends in Computer Vision"**
Seminar at Korea Polytechnic University (now Tech University of Korea)

2021 **"6D Object Pose Estimation"**
Seminar series at CUBOX (now SECERN AI)

PROFESSIONAL SERVICE

2024, 2026 **Reviewer, International Conference on Pattern Recognition (ICPR)**

TECHNICAL SKILLS

Languages & Frameworks: Python, PyTorch, C++, MATLAB; timm, MMDetection, MMSegmentation, ROS

REFERENCES

Dr. Dong Hwan Kim
KIST
gregorykim@kist.re.kr

Prof. Soohwan Kim
Kwangwoon University
kimsoohwan@kw.ac.kr

Dr. Yoonseob Lim
KIST
yslim@kist.re.kr